GymPassJUST code walkthrough

Version 1.2.2 | Read this beside the source code

Read these files first

File	Main purpose
public/index.html	Loads the styles and browser JavaScript
public/ui.js	Builds login, navigation and form HTML
public/app.js	Shows pages, handles clicks and calls the server
public/i18n.js	Switches page text between Arabic and English
src/server.js	Defines URLs and checks who may call each one
src/service.js	Applies the gym rules
src/db.js	Creates SQLite tables
src/local-auth.js	Registers users, checks passwords and creates sessions

Follow one request

1. The student presses Book. The browser posts a class ID to /api/bookings.

2. server.js reads the signed-in user and calls service.book.

3. service.js checks the class, booking window, plan, duplicates and time clashes.

4. It counts occupied seats. A free seat means confirmed; a full class means waitlisted.

5. SQLite saves the booking and notification in one transaction.

6. The result returns to app.js, which updates the page.

A transaction groups related changes. They all succeed together, or none are saved. This prevents two simultaneous bookings from taking the same final seat.

The important service functions

Function	What to explain
membership	Find a plan covering the time and class type
bookedSeats	Count the occupied places in a class
eligible	Check a waiting student has a suitable plan and no clash
book	Confirm or waitlist a valid booking
cancel	Cancel a student's booking and try the next waiter
promote	Go through the waiting list in its saved order
buy	Save a simulated plan purchase and reject active duplicates
cancelClass	Cancel an instructor/manager's class and notify students
makeToken / verifyToken	Sign a short-lived QR and check its signature and dates
scanAttendance	Check instructor, time, QR, subscription and class booking
tick	Finish attendance and record no-shows after class end

gymClass means the selected class, booking means a reservation and subscription means a saved membership. These names replace single-letter variables. The API field names and database columns stay the same.

Follow a QR scan

The student pass contains a signed account reference and an expiry time. It does not contain the student's password. The assigned instructor sends the selected class ID and QR text to /api/class-qr-check-ins.

The checks run in this order: instructor owns the class; class is running; QR is valid; student account exists; a suitable active plan covers the class; and a confirmed booking exists. Success changes the booking to checked_in. A second valid scan returns Already checked in. The minute job later changes it to attended. Unscanned confirmed bookings become no_show only after the class ends.

Explain login simply

local-auth.js stores a salted password hash. On login, it checks the entered password and creates a private session cookie. Later requests use that session to identify the user. The server decides the role; choosing a different role in a browser request does not grant permission.

auth.js also contains the optional university identity-provider path and the isolated test identity flow. They are separate from normal local email/password signup. legacy-data.js upgrades older generated datasets; it is not part of the normal booking flow.

Explain the interface

ui.js contains the shared page HTML. app.js chooses the page and handles forms. classCard builds one timetable card: work out which button is available, choose a status, then return the HTML. escapeText makes user text safe to display. style.css supplies the base styles and theme.css supplies the blue university appearance.